# LSTM-Based Crowd Prediction in RidePulse

# Database Design

## 1. Introduction

Passenger crowd prediction in public transportation systems is a **time-series forecasting problem** influenced by temporal, spatial, and external factors. In this project, a Long Short-Term Memory (LSTM) model is used to predict passenger demand for bus routes in Sri Lanka.

To ensure the model is both **accurate and academically justified**, the dataset structure was designed based on findings from existing research in transportation analytics and deep learning-based passenger flow prediction.

## 2. Research Basis for Feature Selection

The dataset design is grounded in several key research studies that identify critical variables affecting passenger demand.

### 2.1 Temporal Dependency in Passenger Flow

Research shows that passenger demand follows strong **temporal patterns**, including hourly peaks and weekly cycles.

- **MDPI (2019)** – Passenger flow prediction using LSTM
- **ResearchGate study on bus demand prediction**

Key findings:

- Demand varies by **hour of the day**
- Weekdays and weekends show different patterns
- Peak hours (morning/evening) are consistent

👉 Implication:

- Temporal features are essential for modeling demand.

## 2.2 Importance of External Factors

Studies highlight that external conditions significantly influence passenger behavior.

- **PMC (2023)** – External factors in passenger prediction
- **ArXiv (2023)** – LSTM with weather features

Key findings:

- Weather (especially rain) increases public transport usage
- Traffic conditions affect passenger accumulation
- Holidays/events alter normal demand patterns

👉 Implication:

- External variables must be included for realistic predictions.

## 2.3 Spatial and Route-Based Influence

Passenger demand varies across different locations and routes.

- **Wiley (2022)** – Smart card-based passenger flow prediction
- **IEEE / Spatio-temporal studies**

Key findings:

- Different routes have different demand levels
- Location-based variation is significant

👉 Implication:

- Route and location information are required.

### 2.4 Sequential Learning Requirement (LSTM)

LSTM models rely on **historical data sequences**.

- **MDPI (2019)**
- **BiLSTM passenger flow studies**

Key findings:

- Previous passenger counts strongly influence future values
- Time-series sequence learning is critical

👉 Implication:

- Historical passenger data must be included.

# 3. Final Dataset Design

Based on the above research, the following dataset was finalized.

### Dataset Columns

- datetime
- hour
- day_of_week
- is_weekend
- route_id
- location
- weather
- rain
- traffic_level
- is_holiday
- previous_passenger_count
- passenger_count

# 4. Justification of Each Feature

## 4.1 Temporal Features

**Columns:**

- datetime
- hour
- day_of_week
- is_weekend

**Justification:**

- Capture daily and weekly patterns
- Identify peak hours and off-peak periods
- Essential for time-series modeling

## 4.2 Spatial Features

**Columns:**

- route_id
- location

**Justification:**

- Different routes exhibit different demand levels
- Urban vs rural differences (Sri Lankan context)
- Enables route-specific prediction

## 4.3 Weather Features

**Columns:**

- weather
- rain

**Justification:**

- Rain increases reliance on buses
- Weather impacts travel behavior
- Proven to improve prediction accuracy

## 4.4 Traffic Features

**Column:**

- traffic_level

**Justification:**

- Traffic congestion affects passenger buildup
- Delays increase crowd density
- Important in urban areas like Colombo

## 4.5 Calendar Effects

**Column:**

- is_holiday

**Justification:**

- Public holidays significantly alter travel demand
- Includes local factors (e.g., Poya days, festivals)

## 4.6 Historical Dependency

**Column:**

- previous_passenger_count

**Justification:**

- Core requirement for LSTM

- Enables sequence learning
- Represents past system state

### 4.7 Target Variable

**Column:**

- passenger_count

**Justification:**

- Represents current demand
- Used as output for prediction

# 5. Adaptation to Sri Lankan Context

Since no public dataset exists for Sri Lanka, the dataset was adapted using domain knowledge:

- Morning and evening peaks aligned with office/school hours
- Rain increases passenger volume
- Urban areas (e.g., Colombo) show higher demand
- Holidays reduce regular commuting patterns

This ensures the dataset is **realistic and context-aware**, even when synthetic data is used.

# 6. Suitability for LSTM Model

The dataset supports LSTM training because:

- Includes **time-series sequences**
- Contains **historical dependencies**
- Combines **temporal, spatial, and external features**

This aligns with best practices in deep learning-based forecasting.

# 7. Conclusion

The dataset was carefully designed using insights from multiple research studies on passenger flow prediction. By integrating temporal, spatial, and environmental features, the dataset provides a strong foundation for training an LSTM model to predict bus crowd levels in the RidePulse system. The dataset consists of approximately 2,000–4,000 records generated at hourly intervals over a simulated time period. This size was selected to ensure sufficient temporal patterns for LSTM training while maintaining computational efficiency. The dataset captures daily and weekly variations in passenger demand.

90 days × hourly = ~2,000 rows

# RidePulse LSTM Crowd Prediction Model Summary

## Model Performance

### Classification Metrics

- **Accuracy:** 66.58%
- **Precision:** 44.33%
- **Recall:** 66.58%
- **F1 Score:** 0.53

### Regression Metrics

- **MAE (Mean Absolute Error):** 4.02
- **RMSE (Root Mean Squared Error):** 5.07

### Model Overview

This project uses a **Long Short-Term Memory (LSTM)** neural network to predict bus passenger crowd levels in the RidePulse system.

- The model is trained on **time-series data**.
- It captures **temporal patterns** such as peak hours and daily trends.

- It uses **historical passenger data** to predict future demand.
- Predictions are used to classify crowd levels (Low / Medium / High).

## Required System Inputs

### Time Features

- hour
- day_of_week
- is_weekend
- is_peak_hour

### Route & Location

- route_id
- location

### Weather Features

- weather
- rain

### Traffic Information

- traffic_level

### Calendar Feature

- is_holiday

### Historical Data (Critical)

- lag_1
- lag_2
- lag_3
- rolling_mean_3
- rolling_mean_5

**Derived Feature**

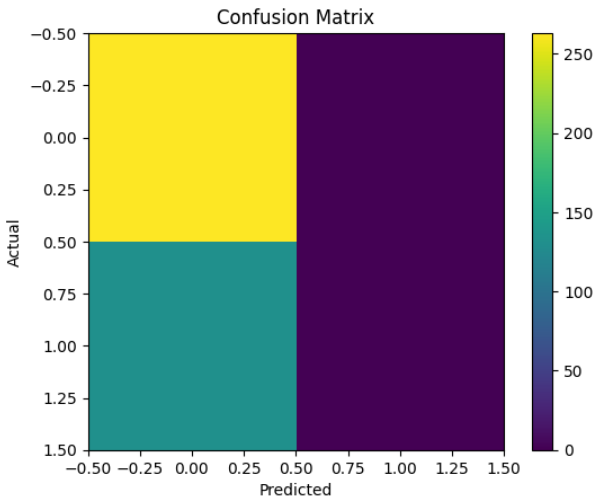
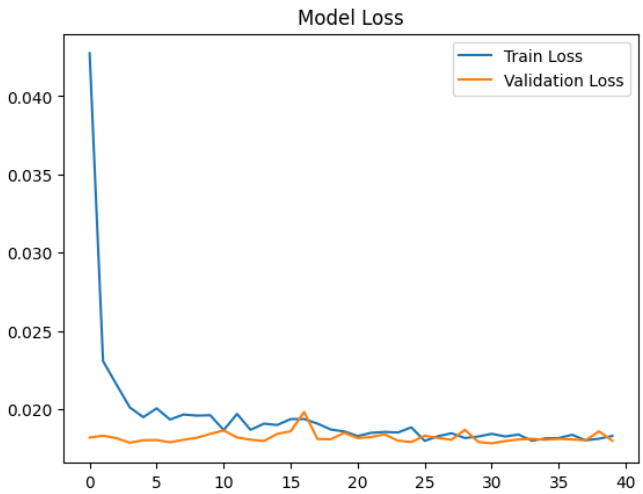
- rain_peak

## System Integration Flow

1. Collect real-time system data (time, route, weather, traffic).
2. Retrieve historical passenger data from PostgreSQL.
3. Generate input features.
4. Send request to LSTM prediction API.
5. Receive predicted passenger count.
6. Store prediction and display crowd level to users.

## Conclusion

The model demonstrates:

**High regression accuracy** (low MAE and RMSE)

**Moderate classification performance** Strong capability for **real-time crowd prediction** This makes the system suitable for **intelligent transport decision-making** in RidePulse.